IST 707: Applied Machine Learning

Course Syllabus

Instructor: Christopher Dunham; cndunham@syr.edu

Class Meetings: Wednesdays, 6:00 PM - 7:30 PM ET

Office Hours: Office hours are held immediately after each class session.

Email: When communicating by email, the subject line should read **IST 707: [Brief description of question/issue]**. If you have an issue with code, please attach your code file to the email and provide a detailed explanation of your issue in the body of the email. Note that I do not typically correspond outside of normal working hours (ET) or on weekends.

Brief Description: An introduction to advanced machine learning techniques and algorithms. Focus on machine learning model building and optimization, real-world applications, communication, ethics, and future directions in the field. Hands-on experience with modern technical tools for machine learning.

Detailed Description: This course will introduce the application of advanced machine learning methods for extracting patterns and knowledge from data to address stakeholder needs. The principles and theories underlying the methods will be discussed, and methods will be related to the issues in applying to real business and research problems. Students will acquire hands-on experience using state-of-the-art tools to apply the methods covered. A central focus in this course is on communication, including translating stakeholder needs into technical methods, and communicating results in a transparent and compelling manner.

The topics of the course will include the key tasks in applied machine learning, including data preparation, concept description, unsupervised and supervised techniques, deep learning, feature construction, optimization, evaluation, analysis, communication, and ethics. Through the exploration of the concepts and techniques of data analytics and practical exercises, students will develop skills that can be applied to business, science, or other organizational problems.

Course Prerequisites: IST 687 OR IST 387 with a minimum grade of B or higher.

NOTE: This is a Python intensive course; students are expected to come with the ability to program in Python and should have some familiarity with Pandas and Matplotlib / Seaborn.

Audience: Graduate students.

Credits: 3

Course Fees and/or Costs: There are no course fees or additional costs beyond tuition that the student should expect to incur.

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Learning Objectives: After taking this course, students will be able to:

1. Document, analyze, and translate stakeholder needs into technical designs and solutions.

2. Select appropriate machine learning techniques for given stakeholder problems and articulate the tradeoffs among different approaches.

3. Optimize machine learning approaches to obtain high-quality solutions to selected problems.

Required Text: *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems* by Aurélien Géron (Third edition, 2022)

Software: Python 3 & associated libraries for machine learning and data analytics; Google Colab.

Note to students: All assignments will be submitted as Jupyter Notebooks and/or GitHub flavored Markdown. No Microsoft Word documents or PDFs will be accepted!

Course Requirements and Expectations:

Attendance/Class flow: This course covers a tremendous amount of material; to make this possible, you are responsible for reviewing all class materials *before* class. Class time will be focused on worked examples and some review and discussion.

Assessment	Percent of Grade
Weekly Participation	20% (10 x 2%)
Homework assignments	40% (5 x 8%)
Project	Total = 40% Proposal – 10% Presentation – 10% Report – 20%

Weekly Participation: Students are required to participate in class discussions, review, and exercises. Review questions and exercises are designed to encourage students to practice their newly learned knowledge, and thus grading is based on participation only, not performance. Students who are not present in class do not earn participation points for that week.

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Homework assignments: Assignments must be prepared using GitHub flavored Markdown and/or Jupyter notebooks submitted electronically to the LMS.

- Pure narrative text (no code) must be submitted as Markdown files (no Word or PDF files will be accepted) and must be free of spelling and grammatical errors.
- Jupyter notebooks will include sections that will explain how each item in the notebook addresses points in the rubric. Notebooks will not contain any broken code and will include results for each runnable cell (i.e., all cells must be run prior to submission).

Group Project: The objective of the project is to apply concepts taught in this class to solve a data analytics problem. **Students may work in pairs or individually**. The project consists of 3 elements:

1. **Project idea proposal**: An overview of the problem, with specific attention paid to the *business case* (i.e., how will your project benefit stakeholders), as well as *why* your problem should be addressed by machine learning, and *how* you plan to do this. Your proposal will include links to and a description of the data set(s) you will use and your proposed machine learning approach.

2. **Project presentation:** A high-level presentation, using a slide deck, presented live during the final class meeting. The presentation should be engaging and succinct, providing a restatement of the problem, the methods used, and a summary of findings. Aim for no more than 5-6 minutes. Each member of the group must be present to receive credit and must contribute equally to the presentation.

3. **Project report:** A comprehensive report due at the end of the term. The report should be no longer than 8 pages + a citations page.

Grading: Below is the formula for number-to-letter grade conversion. *Note: Grades of D and D- may not be assigned to graduate students.*

Grade	Points	Grade	Points	Grade	Points	Grade	Points
A	93-100	B+	87-89	C+	77-79	F	0-69
A-	90-92	B	83-86	C	73-76		
		B-	80-82	C-	70-72		

Tips for success in this class: Curiosity, critical thinking, and programming.

- **Curiosity:** Be curious about the data, pay attention to the data details. Don't treat a data set as a black box. Don't treat an algorithm as a black box. Try to see through them to discover the underlying *meaning* of the data and technologies you employ.
- **Critical thinking:** Using machine learning involves iteration, research, and sense making. You will learn and practice methods to discover patterns and evaluate whether and why the discovered patterns are true and useful.
- **Programming:** You will need some programming ability, and you will develop your skills throughout the course. If you do not have programming experience, you *must* find ways to educate yourself. Numerous resources are available online, and the application of machine learning always requires some dependence upon these resources to fill gaps in your own knowledge.

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iSchool Values: Excellence; Discovery & Innovation; Integrity; Diversity & Inclusion; Global Citizenship and Engagement

Syracuse University has a variety of policies designed to guarantee that students live and study in a community respectful of their needs and those of fellow students. These statements are an official part of this course syllabus.

Academic Integrity Policy: As a pre-eminent and inclusive student-focused research institution, Syracuse University considers academic integrity at the forefront of learning, serving as a core value and guiding pillar of education. Syracuse University's Academic Integrity Policy provides students with the necessary guidelines to complete academic work with integrity throughout their studies. Students are required to uphold both course-specific and university-wide academic integrity expectations such as crediting your sources, doing your own work, communicating honestly, and supporting academic integrity. The full Syracuse University Academic Integrity Policy can be found by visiting class.syr.edu, selecting, "Academic Integrity," and "Expectations and Policy."

Upholding Academic Integrity includes the protection of faculty's intellectual property. Students should not upload, distribute, or share instructors' course materials, including presentations, assignments, exams, or other evaluative materials without permission. Using websites that charge fees or require uploading of course material (e.g., Chegg, Course Hero) to obtain exam solutions or assignments completed by others, which are then presented as your own violates academic integrity expectations in this course and may be classified as a Level 3 violation. All academic integrity expectations that apply to in-person assignments, quizzes, and exams also apply online.

Students found in violation of the policy are subject to grade sanctions determined by the course instructor and non-grade sanctions determined by the School or College where the course is offered. Students may not drop or withdraw from courses in which they face a suspected violation. Any established violation in this course may result in course failure regardless of violation level.

Use of Artificial Intelligence in Course: Based on the specific learning outcomes and assignments in this course, artificial intelligence is permitted on take-home assignments and the final project. See each assignment's instructions for more information about what artificial intelligence tools are permitted and to what extent, as well as citation requirements. If no instructions are provided for a specific assignment, then no use of any artificial intelligence tool is permitted. Any AI use beyond that which is detailed in course assignments is explicitly prohibited except when documented permission is granted.

Academic Integrity Online: Conduct of Online Assessments: All academic integrity expectations that apply to in-person quizzes and exams also apply to online quizzes and exams. In this course, all work submitted for quizzes and exams must be yours alone. Discussing quiz or exam questions with anyone during the quiz or exam period violates academic integrity expectations for this course.

Using websites that charge fees or require uploading of course material (e.g., Chegg, Course Hero) to obtain exam solutions or assignments completed by others and present the work as your own violates academic integrity expectations in this course and may be classified as a Level 3 violation, resulting in suspension or expulsion from Syracuse University

Religious Observances Notification and Policy: Syracuse University's Religious Observances Policy recognizes the diversity of faiths represented in the campus community and protects the rights of students, faculty, and staff to observe religious holy days according to their traditions. Under the policy, students are given an opportunity to make up any examination, study, or work requirements that may be missed due to a religious observance, provided they notify their instructors no later than the academic drop deadline. For observances occurring before the drop deadline, notification is required at least two academic days in advance. Students may enter their observances in MySlice under Student Services/Enrollment/My Religious Observances/Add a Notification.

Disability-Related Accommodations: Syracuse University values diversity and inclusion; we are committed to a climate of mutual respect and full participation. There may be aspects of the instruction or design of this course that result in barriers to your inclusion and full participation in this course. I invite any student to contact me to discuss strategies and/or accommodations (academic adjustments) that may be essential to your success and to collaborate with the Center for Disability Resources (CDR) in this process.

If you would like to discuss disability-accommodations or register with CDR, please visit Center for Disability Resources. Please call (315) 443-4498 or email disabilityresources@syr.edu for more detailed information.

The CDR is responsible for coordinating disability-related academic accommodations and will work with the student to develop an access plan. Since academic accommodations may require early planning and generally is not provided retroactively, please contact CDR as soon as possible to begin this process.

Discrimination or Harassment: Federal and state law, and University policy prohibit discrimination and harassment based on sex or gender (including sexual harassment, sexual assault, domestic/dating violence, stalking, sexual exploitation, and retaliation). If a student has been harassed or assaulted, they can obtain confidential counseling support, 24-hours a day, 7 days a week, from the Sexual and Relationship Violence Response Team at the Counseling Center (315-443-8000, Barnes Center at The Arch, 150 Sims Drive, Syracuse, New York 13244). Incidents of sexual violence or harassment can be reported non-confidentially to the University's Title IX Officer

(Sheila Johnson Willis, 315-443-0211, titleix@syr.edu, 005 Steele Hall). Reports to law enforcement can be made to the University's Department of Public Safety (315-443-2224, 005 Sims Hall), the Syracuse Police Department (511 South State Street, Syracuse, New York, 911 in case of emergency or 315-435-3016 to speak with the Abused Persons Unit), or the State Police (844-845-7269). I will seek to keep information you share with me private to the greatest extent possible, but as a professor I have mandatory reporting responsibilities to share information regarding sexual misconduct, harassment, and crimes I learn about with the University's Title IX Officer to help make our campus a safer place for all.

Health and Wellness: Mental health and overall well-being are significant predictors of academic success. As such it is essential that during your college experience you develop the skills and resources effectively to navigate stress, anxiety, depression, and other mental health concerns. Please familiarize yourself with the range of resources the Barnes Center provides (https://ese.syr.edu/bewell/) and seek out support for mental health concerns as needed. Counseling services are available 24/7, 365 days, at 315-443-8000.

Course Evaluations: At the end of the term, the iSchool will ask you to share course feedback through EvaluationKIT [https://coursefeedback.syr.edu]. Log in to EvaluationKIT using your NetID and password. Please take the time to share your feedback about this course and your experience in it; all ratings and comments are completely anonymous. The iSchool carefully reviews your feedback. Our instructors use this feedback to fine tune course delivery and instruction; our professors of record use this feedback to fine tune course content and assignments. All feedback is factored into iSchool decisions about course, program, and instructor development.

Academic work: Academic work completed during a semester may be used by professors for educational purposes in courses during the semester. Students' registration and continued enrollment constitute consent for this purpose. Before using students' work for educational purposes in subsequent semesters, professors will either request students' permission in writing or render the work anonymous by removing all personal identification.

Course Specific Policies:

- **Registration:** Students must register prior to the first class or may be restricted from registering. If you are registered but not present at the first class, you run the risk of being administratively deregistered from this course so that your seat can be given to a student on the wait list.
- **Class Format & Preparation Before Class:** Lectures for this class are pre-recorded and posted to the LMS. Additionally, there are Jupyter notebooks in each module containing important course content beyond what is presented in the book. Preparation for weekly synchronous class meetings consists of

completing assigned readings, watching pre-recorded lectures, and reviewing any Jupyter notebooks included in that week's module. As part of your engagement with the learning process, it is expected you will prepare any questions or doubts you have about the coursework and come to class ready to discuss the week's material. The in-class session provides time to reinforce concepts through active discussion and hands-on exercises.

- **Attendance:** Students are expected to attend all class meetings and be prepared to participate in class discussion.
- **Late Policy for Assignments:** Any assignment handed in after the deadline (not counting a 1-hour grace period) is automatically discounted by 1 point of the final grade per day late.

Communications: This course will use the 2U Digital Campus as the main communication platforms for class exercises and notifications. Students are required to check their 2U Digital Campus accounts and email on a regular basis.

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Course Schedule

Unless otherwise specified, homework assignments are due by 11:59 PM ET, Sunday of the week they are due. All readings are from the Géron textbook, 3rd edition. Readings should be completed before class in the week they are assigned.

Week	Topics	Readings	Events
01 Oct 02	Introduction to Machine Learning & Machine Learning Ecosystem	Chapter 1	
02 Oct 09	Machine Learning Workflow	Chapter 2	
03 Oct 16	Data Preparation & Evaluation	Chapter 3	HW 1 due (Sun Oct 20)
04 Oct 23	Classifiers I: Logistic Regression; KNN; SVM	Chapters 4 & 5	HW 2 due (Sun Oct 27)
05 Oct 30	Classifiers II: Decision Trees; Ensemble Methods	Chapters 6 & 7	
06 Nov 06	Dimensionality Reduction	Chapter 8	HW 3 due (Sun Nov 10)
07 Nov 13	Clustering	Chapter 9	Project Proposal Due
08 Nov 20	Deep Learning: Introduction	Chapter 10	HW 4 due (Sun Nov 24)
09 Nov 27	Deep Learning: Activation Functions; Keras Functional API; Tensorflow Tools	Chapter 11	HW 5 due (Sun Dec 1)
10 Dec 04	Deep Learning: Learning Schedules; Optimizers; Regularization; Hyperparameter Tuning		
11 Dec 11	Project presentations Final reports due last day of term		